CSSMedia Complete Reference



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Complete reference

Look up every selector, property, value and at-rule by subject.

Each table links to the CSS specification that defines its entries. The chapters of this guide end with a shorter reference that lists the entries they explain. The complete references are grouped by subject:

- Selector reference
- Font reference
- Box model properties
- Dimension properties
- Background and image reference
- Display and positioning
- Page reference
- Links and generated content reference
- Footnotes
- Links
- Named strings
- Cascade reference

Selector reference

These tables list every selector that CSSMedia accepts. The `In a PDF` column says whether the selector matches elements while CSSMedia generates a PDF.

Basic selectors

Basic selectors match an element by its name, ID or attributes. See Basic selectors and namespaces.

Name	Description	Values	In a PDF	Specification
`E`	Type selector. Matches elements named `E`.	An element name.	Matches	Selectors 4 §5.1
`*`	Universal selector. Matches any element.		Matches	Selectors 4 §5.2
`#id`	ID selector. Matches the element whose `id` or `xml:id` attribute has this value.	An identifier.	Matches	Selectors 4 §6.7
`.name`	Class selector. Matches inside SVG and MathML content, which define a `class` attribute.	An identifier.	Matches in SVG and MathML only	Selectors 4 §6.6
`[attr]`	Matches elements that have the attribute.	An attribute name, with an optional namespace prefix.	Matches	Selectors 4 §6
`[attr=value]`	Matches when the attribute value is exactly `value`.	A string or an identifier. Add `i` to ignore ASCII case or `s` to require it.	Matches	Selectors 4 §6.1
`[attr~=value]`	Matches when the value is one of the white-space separated words of the attribute.	A word without white space.	Matches	Selectors 4 §6.1
`[attr|=value]`	Matches when the attribute is `value` or starts with `value-`.	A string or an identifier.	Matches	Selectors 4 §6.1

Name	Description	Values	In a PDF	Specification
`[attr^=value]`	Matches when the attribute starts with `value`.	A non-empty string.	Matches	Selectors 4 §6.2
`[attr$=value]`	Matches when the attribute ends with `value`.	A non-empty string.	Matches	Selectors 4 §6.2
`[attr*=value]`	Matches when the attribute contains `value`.	A non-empty string.	Matches	Selectors 4 §6.2
`ns|E`, `*|E`, `|E`	Limits a type selector to a declared namespace, to any namespace, or to no namespace. The same prefixes work in attribute selectors.	A prefix declared with `@namespace`.	Matches	Selectors 4 §5.3, CSS Namespaces 3 §3
`&`	Nesting selector. Stands for the selector of the parent rule.		Matches	CSS Nesting 1

Combinators

A combinator relates two selectors. See Combinators.

Name	Description	Values	In a PDF	Specification
`A B`	Descendant combinator. Matches `B` inside `A` at any depth.	Two selectors separated by white space.	Matches	Selectors 4 §14.1
`A > B`	Child combinator. Matches `B` when its parent is `A`.	Two selectors.	Matches	Selectors 4 §14.2
`A + B`	Next-sibling combinator. Matches `B` when it directly follows `A`.	Two selectors.	Matches	Selectors 4 §14.3
`A ~ B`	Subsequent-sibling combinator. Matches `B` when `A` is an earlier sibling.	Two selectors.	Matches	Selectors 4 §14.4
`A || B`	Column combinator. Matches the cells `B` of the column that `A` describes.	Two selectors.	Matches	Selectors 5 §8.1

Name	Description	Values	In a PDF	Specification
`A / attr/ B`	Reference combinator. Matches B when the attribute `attr` of A refers to its ID.	Two selectors and an attribute name between slashes.	Matches	Selectors 5 §9.1

Pseudo-classes

A pseudo-class tests a property of an element other than its name and attributes. See Pseudo-classes.

Name	Description	
`:root`	The root element of the document.	
`:empty`	An element with no child elements and no text other than white space.	
`:scope`	The scoping element. Without another scope, the root element.	
`:first-child`, `:last-child`, `:only-child`	The first, last or only element among its siblings.	
`:first-of-type`, `:last-of-type`, `:only-of-type`	The first, last or only element of its type among its siblings.	
`:nth-child()`, `:nth-last-child()`	Elements at a position among their siblings, counted from the start or from the end.	odd, An+B follo and list.
`:nth-of-type()`, `:nth-last-of-type()`	Elements at a position among the siblings of the same type.	odd, An+B

Name	Description	
`:nth-col()`, `:nth-last-col()`	Table cells at a position among the columns of their table.	odd, An+B
`:is()`, `:matches()`	An element that matches any selector of the list. `:matches()` is the old name of `:is()`.	A sel Inval are i
`:where()`	Works as `:is()` and adds no specificity.	A sel Inval are i
`:not()`	An element that matches none of the selectors of the list.	A sel An in selec the v selec
`:has()`	An element that has a related element matching the selector.	A rel selec such `figu` cann anot or a elem
`:lang()`	Elements in a language. The language comes from `xml:lang` or the `-cssmedia-lang` property.	One lang rang iden strin
`:dir()`	Elements by text direction, read from the text and the language, not from the `direction` property.	`ltr`

Name	Description	
`:link`, `:any-link`	Elements that are hyperlinks, because `-cssmedia-link` gives them a destination.	
`:local-link`	Hyperlinks that point to the same document. The optional number limits how many path segments are compared.	An o nega in pa
`:read-only`, `:defined`	Every element is read-only and defined, because an XML element has no editable or undefined state.	
`:hover`, `:active`, `:focus`, `:focus-visible`, `:focus-within`	User action states, such as pointing at an element or giving it keyboard focus.	
`:interest-source`, `:interest-target`	Source and target of a user's declared interest.	
`:visited`, `:target`	Links that were visited, and the element that a link points to. A PDF has no browsing history and no current target.	
`:playing`, `:paused`, `:seeking`, `:buffering`, `:stalled`, `:muted`, `:volume-locked`	Media playback, loading and sound states.	

Name	Description	
`:current`, `:current()`, `:past`, `:future`	Elements in relation to a timeline, such as speech or video playback.	A sel for :
`:open`, `:popover-open`, `:modal`, `:fullscreen`, `:picture-in-picture`	Display states of interactive elements.	
`:enabled`, `:disabled`, `:read-write`, `:placeholder-shown`, `:autofill`, `:default`, `:checked`, `:unchecked`, `:indeterminate`, `:blank`	Form control states. An XML element is never a form control.	
`:valid`, `:invalid`, `:in-range`, `:out-of-range`, `:required`, `:optional`, `:user-valid`, `:user-invalid`	Form validation states.	
`:state()`	A custom state that a document language exposes. XML has none.	An id
`:heading`, `:heading()`	Headings, as a document language defines them. CSSMedia does not give XML elements a heading role.	A list for :

Pseudo-elements

A pseudo-element selects a part of an element or generated content. See Pseudo-elements.

Name	Description	Values	In a PDF	Specification
`::before`, `::after`	Generated content before or after the content of an element. The `content` property gives it.		Matches	CSS Pseudo-Elements 4 §4.1
`::marker`	The marker of a list item. Also `::before::marker` and `::after::marker`.		Matches	CSS Pseudo-Elements 4 §4.2

Name	Description	Values	In a PDF	Specification
`::first-line`	The first formatted line of a block.		Matches	CSS Pseudo-Elements 4 §2.1
`::first-letter`	The first letter of a block, with the punctuation around it. `::first-letter::prefix` and `::first-letter::suffix` select that punctuation.		Matches	CSS Pseudo-Elements 4 §2.2
`::footnote-call`	The reference that replaces a footnote in the text. By default it is the value of the `footnote` counter, as a superscript.		Matches	CSS GCPM 3 §2.6
`::footnote-marker`	The marker at the start of a footnote body. By default it is the value of the `footnote` counter, followed by a period and a space.		Matches	CSS GCPM 3 §2.7
`::selection`, `::search-text`, `::target-text`, `::spelling-error`, `::grammar-error`, `::highlight()`	Highlighted ranges created by selection, search, navigation, spelling and grammar checking, or scripts.	A custom identifier for `::highlight()`.	No range exists	CSS Pseudo-Elements 4 §3.1
`::placeholder`, `::file-selector-button`, `::details-content`	Parts of form controls and disclosure widgets.		No such part exists	CSS Pseudo-Elements 4 §4.3, §5.1, §5.2
`::backdrop`	The backdrop of an element shown in the top layer.		No such part exists	CSS Position 4 §3.2

Font reference

This table covers the font properties, at-rules, and descriptors introduced in the font guides.

Name	Description	Values	Default	Specification
`font`	Sets the font family, size, style, weight, width, variant, and line height in one declaration.	A system-font keyword (`caption`, `icon`, `menu`, `message-box`, `small-caption`, or `status-bar`), or optional font-style, `normal` or `small-caps`, font-weight, and font-width values followed by required font size and family and optional line height		CSS Fonts 4, `font`
`font-family`	Selects a font family or ordered list of fallback families.	Family names and `serif`, `sans-serif`, `system-ui`, `cursive`, `fantasy`, `math`, `monospace`, `ui-serif`, `ui-sans-serif`, `ui-monospace`, `ui-rounded`, or `generic(fangsong)`, `generic(kai)`, `generic(khmer-mul)`, `generic(nastaliq)`		CSS Fonts 4, `font-family`
`font-size`	Sets the font size.	`xx-small`, `x-small`, `small`, `medium`, `large`, `x-large`, `xx-large`, `xxx-large`, `larger`, `smaller`, a non-negative length or percentage, or `math`		CSS Fonts 4, `font-size`
`font-size-adjust`	Adjusts font sizing using a selected font metric.	none, optional metric (`ex-height`, `cap-height`, `ch-width`, `ic-width`, or `ic-height`), then `from-font` or a non-negative number		CSS Fonts 5, `font-size-adjust`
`font-weight`	Selects the requested weight of a font face.	`normal`, `bold`, `bolder`, `lighter`, or a number from 1 to 1000		CSS Fonts 4, `font-weight`

Name	Description	Values	Default	Specification
`font-width`	Selects the requested width of a font face.	`normal`, a non-negative percentage, `ultra-condensed`, `extra-condensed`, `condensed`, `semi-condensed`, `semi-expanded`, `expanded`, `extra-expanded`, or `ultra-expanded`		CSS Fonts 4, `font-width`
`font-stretch`	Legacy alias for `font-width`.	`normal`, a non-negative percentage, `ultra-condensed`, `extra-condensed`, `condensed`, `semi-condensed`, `semi-expanded`, `expanded`, `extra-expanded`, or `ultra-expanded`		CSS Fonts 4, `font-stretch`
`font-style`	Selects the requested upright or slanted face.	`normal`, `italic`, `left`, `right`, or `oblique` with an optional angle from -90deg to 90deg		CSS Fonts 4, `font-style`
`font-synthesis`	Controls synthetic weight, style, small-cap, and position forms.	none, or any combination of `weight`, `style`, `small-caps`, and `position`		CSS Fonts 4, `font-synthesis`
`font-synthesis-weight`	Controls synthetic bolding.	`auto` or none		CSS Fonts 4, `font-synthesis-weight`
`font-synthesis-style`	Controls synthetic italic or oblique styling.	`auto`, none, or `oblique-only`		CSS Fonts 4, `font-synthesis-style`
`font-synthesis-small-caps`	Controls synthetic small capitals.	`auto` or none		CSS Fonts 4, `font-synthesis-small-caps`

Name	Description	Values	Default	Specification
`font-synthesis-position`	Controls synthetic subscript and superscript forms.	`auto` or none		CSS Fonts 4, `font-synthesis-position`
`font-variant`	Shorthand for typographic variant properties.	`normal`, none, or combinations of variant longhand values		CSS Fonts 4, `font-variant`
`font-variant-alternates`	Requests named alternate glyphs and stylistic forms.	`normal` or combinations of `stylistic()`, `historical-forms`, `styleset()`, `character-variant()`, `swash()`, `ornaments()`, and `annotation()`		CSS Fonts 4, `font-variant-alternates`
`font-variant-caps`	Requests small-cap and other capitalization forms.	`normal`, `small-caps`, `all-small-caps`, `petite-caps`, `all-petite-caps`, `unicase`, or `titling-caps`		CSS Fonts 4, `font-variant-caps`
`font-variant-east-asian`	Requests East Asian glyph variants and widths.	`normal` or combinations of `jis78`, `jis83`, `jis90`, `jis04`, `simplified`, `traditional`, `full-width`, `proportional-width`, and `ruby`		CSS Fonts 4, `font-variant-east-asian`
`font-variant-emoji`	Requests text or emoji presentation for supported characters.	`normal`, `text`, `emoji`, or `unicode`		CSS Fonts 4, `font-variant-emoji`

Name	Description	Values	Default	Specification
`font-variant-ligatures`	Controls common, discretionary, historical, and contextual ligatures.	`normal`, none, or combinations of `common-ligatures`/`no-common-ligatures`, `discretionary-ligatures`/`no-discretionary-ligatures`, `historical-ligatures`/`no-historical-ligatures`, and `contextual`/`no-contextual`		CSS Fonts 4, `font-variant-ligatures`
`font-variant-numeric`	Requests number shapes, spacing, and fractions.	`normal` or combinations of `lining-nums`/`oldstyle-nums`, `proportional-nums`/`tabular-nums`, `diagonal-fractions`/`stacked-fractions`, `ordinal`, and `slashed-zero`		CSS Fonts 4, `font-variant-numeric`
`font-variant-position`	Requests subscript or superscript glyph forms.	`normal`, `sub`, or `super`		CSS Fonts 4, `font-variant-position`
`font-kerning`	Controls use of kerning information in the font.	`auto`, `normal`, or none		CSS Fonts 4, `font-kerning`
`font-feature-settings`	Sets low-level OpenType feature tags and values.	`normal` or a comma-separated list of feature tags, optionally followed by a non-negative integer, on, or `off`		CSS Fonts 4, `font-feature-settings`
`font-language-override`	Sets the OpenType language system tag.	`normal` or a string		CSS Fonts 4, `font-language-override`

Name	Description	Values	Default	Specification
`font-optical-sizing`	Enables or disables automatic optical-size axis selection for variable fonts.	`auto` or none		CSS Fonts 4, `font-optical-sizing`
`font-variation-settings`	Sets low-level variable-font axis tags and values.	`normal` or a comma-separated list of four-character tags and numbers		CSS Fonts 4, `font-variation-settings`
`font-palette`	Chooses a color palette for a color font.	`normal`, `light`, `dark`, a palette identifier, or `palette-mix()`		CSS Fonts 4, `font-palette`
`line-height`	Sets the preferred height of inline content used to determine line-box height.	`normal`, a non-negative number, non-negative length, or non-negative percentage		CSS Inline 3, `line-height`
`@font-face`	Declares a font face and its source for use by a family name.	Font-face rule with descriptors		CSS Fonts 4, `@font-face`
`@font-face font-family`	Names the family for this face.	A family name		CSS Fonts 4, `font-family` descriptor
`@font-face src`	Lists font resources or installed font names to try in order.	Comma-separated `url()` or `local()` sources with optional format and technology hints		CSS Fonts 4, `src` descriptor
`@font-face font-display`	Sets the fallback and swap behavior while a font loads.	`auto`, `block`, `swap`, `fallback`, or `optional`		CSS Fonts 4, `font-display` descriptor

Name	Description	Values	Default	Specification
`@font-face font-weight`	Describes the weight or weight range supplied by the face.	`auto`, or one or two values, each `normal`, `bold`, or a number from 1 to 1000		CSS Fonts 4, `font-weight` descriptor
`@font-face font-width`	Describes the width or width range supplied by the face.	`auto`, or one or two `font-width` values, each `normal`, a width keyword, or a non-negative percentage		CSS Fonts 4, `font-width` descriptor
`@font-face font-stretch`	Legacy alias for the `font-width` descriptor.	`auto`, or one or two values, each `normal`, a non-negative percentage, `ultra-condensed`, `extra-condensed`, `condensed`, `semi-condensed`, `semi-expanded`, `expanded`, `extra-expanded`, or `ultra-expanded`		CSS Fonts 4, `font-stretch` descriptor
`@font-face font-style`	Describes the style or style range supplied by the face.	`auto`, `normal`, `italic`, `left`, `right`, or `oblique` with no angle or one or two angles from -90deg to 90deg		CSS Fonts 4, `font-style` descriptor
`@font-face unicode-range`	Limits which character code points this face can serve.	One or more Unicode ranges		CSS Fonts 4, `unicode-range` descriptor
`@font-face font-feature-settings`	Sets default low-level OpenType feature settings for a face.	`normal` or a comma-separated list of feature tags, optionally followed by a non-negative integer, on, or `off`		CSS Fonts 4, `font-feature-settings` descriptor
`@font-face font-language-override`	Sets the default OpenType language system tag for a face.	`normal` or a string		CSS Fonts 4, `font-language-override` descriptor

Name	Description	Values	Default	Specification
`@font-face font-named-instance`	Selects a named variation instance in the font.	`auto` or a string		CSS Fonts 4, `font-named-instance` descriptor
`@font-face font-variation-settings`	Sets the face's default variable-font axes.	`normal` or a list of axis tags and numbers		CSS Fonts 4, `font-variation-settings` descriptor
`@font-face font-size`	Describes the face's optical-size value or range.	`auto` or one or two numbers		CSS Fonts 5, `font-size` descriptor
`@font-face ascent-override`	Overrides the face's ascent metric.	One or two values, each `normal` or a non-negative percentage		CSS Fonts 5, `ascent-override`
`@font-face descent-override`	Overrides the face's descent metric.	One or two values, each `normal` or a non-negative percentage		CSS Fonts 5, `descent-override`
`@font-face line-gap-override`	Overrides the face's line-gap metric.	One or two values, each `normal` or a non-negative percentage		CSS Fonts 5, `line-gap-override`
`@font-face size-adjust`	Scales the face's glyph outlines and metrics.	Non-negative percentage		CSS Fonts 5, `size-adjust`
`@font-face subscript-position-override`	Overrides the subscript position metric.	One or two values, each `normal`, `from-font`, or a percentage		CSS Fonts 5, `subscript-position-override`
`@font-face subscript-size-override`	Overrides the subscript size metric.	One or two values, each `normal`, `from-font`, or a non-negative percentage		CSS Fonts 5, `subscript-size-override`
`@font-face superscript-position-override`	Overrides the superscript position metric.	One or two values, each `normal`, `from-font`, or a percentage		CSS Fonts 5, `superscript-position-override`

Name	Description	Values	Default	Specification
`@font-face superscript-size-override`	Overrides the superscript size metric.	One or two values, each `normal`, `from-font`, or a non-negative percentage		CSS Fonts 5, `superscript-size-override`
`@font-feature-values`	Names feature values for one or more font families.	Rule with one or more family names and feature-value blocks		CSS Fonts 4, `@font-feature-values`
`@font-feature-values font-display`	Sets a default font-display policy for the listed families.	`auto`, `block`, `swap`, `fallback`, or `optional`		CSS Fonts 4, `font-display` descriptor
`@font-feature-values @annotation`	Maps names to annotation feature indexes.	Named values mapped to one non-negative integer		CSS Fonts 4, `@annotation`
`@font-feature-values @character-variant`	Maps names to character-variant feature indexes.	One or two integers. Values 1 to 99 select `cv01` to `cv99`; in a two-integer value, the second integer sets that feature's value. Zero and values above 99 parse but select no feature.		CSS Fonts 4, `@character-variant`
`@font-feature-values @historical-forms`	Maps names to historical-forms feature indexes.	Named values mapped to one or more non-negative integers		CSS Fonts 4, `@historical-forms`
`@font-feature-values @ornaments`	Maps names to ornaments feature indexes.	Named values mapped to one non-negative integer		CSS Fonts 4, `@ornaments`
`@font-feature-values @styleset`	Maps names to styleset feature indexes.	Named values mapped to one or more integers; 1 through 99 select `ss01` through `ss99`, while 0 and values above 99 select no OpenType feature		CSS Fonts 4, `@styleset`

Name	Description	Values	Default	Specification
`@font-feature-values @stylistic`	Maps names to stylistic feature indexes.	Named values mapped to one non-negative integer		CSS Fonts 4, `@stylistic`
`@font-feature-values @swash`	Maps names to swash feature indexes.	Named values mapped to one non-negative integer		CSS Fonts 4, `@swash`
`@font-palette-values`	Defines a named palette for a color font.	Rule with a dashed identifier and palette descriptors		CSS Fonts 4, `@font-palette-values`
`@font-palette-values font-family`	Required descriptor that identifies the font family to which the named palette applies.	One or more family names		CSS Fonts 4, `font-family` descriptor
`@font-palette-values base-palette`	Chooses the base palette for the named palette.	`light`, `dark`, or a non-negative integer		CSS Fonts 4, `base-palette`
`@font-palette-values override-colors`	Replaces selected colors in the base palette.	One or more palette index and color pairs		CSS Fonts 4, `override-colors`

Box model properties

This table lists the box model properties described in Box sizing, margins and padding. A length is a number with a unit, such as `12pt` or `1em`. Margin and padding percentages are relative to the containing block's logical width, or inline size.

These properties set the margins, padding and box sizing of the boxes of a document.

The CSS-wide keywords `inherit`, `initial`, `revert`, `revert-layer`, and `unset` can also be used with these properties.

Name	Description	Values	Default	Specification
`box-sizing`	Chooses whether `width` and `height` measure the content area or the box including padding and border.	`content-box` measures the content; `border-box` includes padding and border.		CSS Sizing 3
`margin`	Sets the space outside all four sides of a box.	One to four lengths or percentages, `auto`, or `anchor-size()`.		CSS Box Model 4
`margin-block`	Sets the margins at the start and end of the block axis.	One or two lengths, percentages, `auto`, or `anchor-size()` values.		CSS Logical Properties 1
`margin-block-end`	Sets the margin at the end of the block axis.	Length, percentage, `auto`, or `anchor-size()`.		CSS Logical Properties 1
`margin-block-start`	Sets the margin at the start of the block axis.	Length, percentage, `auto`, or `anchor-size()`.		CSS Logical Properties 1
`margin-bottom`	Sets the space outside the bottom side of a box.	Length, percentage, `auto`, or `anchor-size()`.		CSS Box Model 4
`margin-break`	Controls whether a margin is kept or discarded when its box is split by a page break.	`auto` follows the break rules; `keep` keeps the margin; `discard` removes it.		CSS Fragmentation 4
`margin-inline`	Sets the margins at the start and end of the inline axis.	One or two lengths, percentages, `auto`, or `anchor-size()` values.		CSS Logical Properties 1
`margin-inline-end`	Sets the margin at the end of the inline axis.	Length, percentage, `auto`, or `anchor-size()`.		CSS Logical Properties 1
`margin-inline-start`	Sets the margin at the start of the inline axis.	Length, percentage, `auto`, or `anchor-size()`.		CSS Logical Properties 1
`margin-left`	Sets the space outside the left side of a box.	Length, percentage, `auto`, or `anchor-size()`.		CSS Box Model 4

Name	Description	Values	Default	Specification
`margin-right`	Sets the space outside the right side of a box.	Length, percentage, `auto`, or `anchor-size()`.		CSS Box Model 4
`margin-top`	Sets the space outside the top side of a box.	Length, percentage, `auto`, or `anchor-size()`.		CSS Box Model 4
`margin-trim`	Removes selected margins where child boxes meet the edges of their container.	none; `block`, `inline`, or both; or one or more of `block-start`, `inline-start`, `block-end`, and `inline-end`.		CSS Box Model 4
`padding`	Sets the space between the content and all four sides of its border.	One to four non-negative lengths or percentages.		CSS Box Model 4
`padding-block`	Sets the padding at the start and end of the block axis.	One or two non-negative lengths or percentages.		CSS Logical Properties 1
`padding-block-end`	Sets the padding at the end of the block axis.	Non-negative length or percentage.		CSS Logical Properties 1
`padding-block-start`	Sets the padding at the start of the block axis.	Non-negative length or percentage.		CSS Logical Properties 1
`padding-bottom`	Sets the space between the content and the bottom border.	Non-negative length or percentage.		CSS Box Model 4
`padding-inline`	Sets the padding at the start and end of the inline axis.	One or two non-negative lengths or percentages.		CSS Logical Properties 1
`padding-inline-end`	Sets the padding at the end of the inline axis.	Non-negative length or percentage.		CSS Logical Properties 1
`padding-inline-start`	Sets the padding at the start of the inline axis.	Non-negative length or percentage.		CSS Logical Properties 1
`padding-left`	Sets the space between the content and the left border.	Non-negative length or percentage.		CSS Box Model 4

Name	Description	Values	Default	Specification
`padding-right`	Sets the space between the content and the right border.	Non-negative length or percentage.		CSS Box Model 4
`padding-top`	Sets the space between the content and the top border.	Non-negative length or percentage.		CSS Box Model 4

Dimension properties

This reference lists the CSS sizing properties documented in Box dimensions. The `Values` column gives CSS syntax.

CSS Anchor Positioning permits `anchor-size()` only on properties accepted in an `@position-try` rule. These include sizing, inset, and margin properties. This page lists the sizing properties.

Preferred sizes

Name	Description	Values	Default	Specification
`width`	Sets the preferred physical width.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content(<length-percentage [0,∞]>)`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`height`	Sets the preferred physical height.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content(<length-percentage [0,∞]>)`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`inline-size`	Sets the preferred size on the inline axis.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content()`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`block-size`	Sets the preferred size on the block axis.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content()`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3

- `<length-percentage>` means a length such as `12pt`, a percentage such as `60%`, or a `calc()` expression combining them.

- `[0,∞]` in `fit-content(<length-percentage [0,∞]>)` means the length or percentage cannot be negative.
- `anchor-size()` accepts an optional anchor name, an optional size keyword, or both, followed by an optional length or percentage fallback. It resolves to the anchor's size only when the box is absolutely positioned and has a target anchor. Otherwise, it resolves to the fallback. Without a fallback, the declaration becomes invalid at computed-value time. The sizing properties listed here include physical and logical properties.

Minimum and maximum sizes

Name	Description	Values	Default	Specification
`min-width`	Sets a lower limit for the physical width.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content(<length-percentage [0,∞]>)`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`min-height`	Sets a lower limit for the physical height.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content(<length-percentage [0,∞]>)`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`min-inline-size`	Sets a lower limit on the inline axis.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content()`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`min-block-size`	Sets a lower limit on the block axis.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content()`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`max-width`	Sets an upper limit for the physical width.	`none`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content(<length-percentage [0,∞]>)`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3

Name	Description	Values	Default	Specification
`max-height`	Sets an upper limit for the physical height.	`none`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content(<length-percentage [0,∞]>)`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`max-inline-size`	Sets an upper limit on the inline axis.	`none`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content()`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`max-block-size`	Sets an upper limit on the block axis.	`none`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content()`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3

- `<length-percentage>` means a length such as `12pt`, a percentage such as `60%`, or a `calc()` expression combining them.
- `[0,∞]` in `fit-content(<length-percentage [0,∞]>)` means the length or percentage cannot be negative.

Paired size shorthands

Name	Description	Values	Default	Specification
`size`	Sets `width` and `height`.	One or two preferred-size values.		CSS Sizing 4
`min-size`	Sets `min-width` and `min-height`.	One or two minimum-size values.		CSS Sizing 4
`max-size`	Sets `max-width` and `max-height`.	One or two maximum-size values.		CSS Sizing 4

- Each pair takes the corresponding longhand's values in width-then-height order. The second value is optional and defaults to the first.
- The `size` shorthand is not valid in an `@page` context. That context has a separate `size` descriptor for page dimensions.

Aspect ratio and intrinsic sizes

Name	Description	Values	Default	Specification
`aspect-ratio`	Supplies a preferred ratio between the width and height.	`auto`, a non-negative number such as `1.5`, or two non-negative numbers such as `16 / 9`; `auto` can be combined with a number.		CSS Sizing 4
`contain-intrinsic-width`	Fallback intrinsic width under size containment.	Optional `auto`; none or a non-negative length.		CSS Sizing 4
`contain-intrinsic-height`	Fallback intrinsic height under size containment.	Optional `auto`; none or a non-negative length.		CSS Sizing 4
`contain-intrinsic-inline-size`	Fallback intrinsic inline size under size containment.	Optional `auto`; none or a non-negative length.		CSS Sizing 4
`contain-intrinsic-block-size`	Fallback intrinsic block size under size containment.	Optional `auto`; none or a non-negative length.		CSS Sizing 4
`contain-intrinsic-size`	Sets fallback intrinsic width and height.	One or two longhand values; see below.		CSS Sizing 4

- For `aspect-ratio`, the first number is the width and the second is the height. If you give one number, the second number is `1`. For example, `1.5` means the same ratio as `1.5 / 1`.
- Each `contain-intrinsic-*` longhand accepts an optional `auto`, followed by none or a non-negative length.
- The `contain-intrinsic-size` shorthand takes one or two such values. One value applies to both dimensions.

Replaced elements and intrinsic contributions

Name	Description	Values	Default	Specification
`frame-sizing`	Selects a replaced element's content size for sizing.	`auto` or a `content-*` dimension.		CSS Sizing 4
`min-intrinsic-sizing`	Controls min-content contribution compression.	`legacy`, `zero-if-scroll`, `zero-if-extrinsic`, or both.		CSS Sizing 4

- `frame-sizing` accepts `auto`, `content-width`, `content-height`, `content-block-size`, or `content-inline-size`.

Background and image reference

Name	Description	Values	Default	Specification
`background`	Sets the background color and one or more background image layers. The color, if present, belongs to the final layer.	Comma-separated layers with image, position and optional size, repeat, attachment, and clipping or positioning box values; optional color on the final layer	See individual properties	CSS Backgrounds 4, `background`
`background-attachment`	Selects whether a background image is attached to its box, its contents, or the page box in paged media.	`scroll`, `fixed`, or `local`; comma-separated list	`fixed`	CSS Backgrounds 4, `background-attachment`
`background-blend-mode`	Sets how each background image layer blends with the layers below it and the background color.	`normal`, `darken`, `multiply`, `color-burn`, `lighten`, `screen`, `color-dodge`, `overlay`, `soft-light`, `hard-light`, `difference`, `exclusion`, hue, `saturation`, `color`, or `luminosity`; comma-separated list	`normal`	Compositing and Blending 1, `background-blend-mode`
`background-clip`	Sets the box area where the background is painted.	`border-box`, `padding-box`, `content-box`, `border-area`, `text`, or both `border-area` and `text`; comma-separated list	`border-box`	CSS Backgrounds 4, `background-clip`

Name	Description	Values	Default	Specification
`background-color`	Sets the background color.	`<color>`	`transparent`	CSS Backgrounds 4, `background-color`
`background-image`	Sets one or more background images or gradients.	One or more comma-separated none or `<image>` values	`none`	CSS Backgrounds 4, `background-image`
`background-origin`	Sets the box used as the positioning area for each background image.	`border-box`, `padding-box`, or `content-box`; comma-separated list	`padding-box`	CSS Backgrounds 4, `background-origin`
`background-position`	Sets the position of each background image.	One- to four-part position using keywords, lengths, or percentages; comma-separated list	`0% 0%`	CSS Backgrounds 4, `background-position`
`background-position-block`	Sets the position of each background image on the block axis.	`center`, or `start` or end, optionally combined with a length or percentage; comma-separated list	`0%`	CSS Backgrounds 4, `background-position-block`
`background-position-inline`	Sets the position of each background image on the inline axis.	`center`, or `start` or end, optionally combined with a length or percentage; comma-separated list	`0%`	CSS Backgrounds 4, `background-position-inline`
`background-position-x`	Sets the position of each background image on the physical horizontal axis.	`center`; `left`, `right`, `x-start`, or `x-end`; a length or percentage; or a side keyword with a length or percentage; comma-separated list	`0%`	CSS Backgrounds 4, `background-position-x`

Name	Description	Values	Default	Specification
`background-position-y`	Sets the position of each background image on the physical vertical axis.	`center`; `top`, `bottom`, `y-start`, or `y-end`; a length or percentage; or a side keyword with a length or percentage; comma-separated list	`0%`	CSS Backgrounds 4, `background-position-y`
`background-repeat`	Sets how each background image repeats along both axes.	`repeat`, `space`, `round`, or `no-repeat`; optional second value; `repeat-x`, `repeat-y`, `repeat-block`, or `repeat-inline`; comma-separated list	`repeat`	CSS Backgrounds 4, `background-repeat`
`background-repeat-block`	Sets how each background image repeats on the block axis.	`repeat`, `space`, `round`, or `no-repeat`; comma-separated list	`repeat`	CSS Backgrounds 4, `background-repeat-block`
`background-repeat-inline`	Sets how each background image repeats on the inline axis.	`repeat`, `space`, `round`, or `no-repeat`; comma-separated list	`repeat`	CSS Backgrounds 4, `background-repeat-inline`
`background-repeat-x`	Sets how each background image repeats on the physical horizontal axis.	`repeat`, `space`, `round`, or `no-repeat`; comma-separated list	`repeat`	CSS Backgrounds 4, `background-repeat-x`
`background-repeat-y`	Sets how each background image repeats on the physical vertical axis.	`repeat`, `space`, `round`, or `no-repeat`; comma-separated list	`repeat`	CSS Backgrounds 4, `background-repeat-y`

Name	Description	Values	Default	Specification
`background-size`	Sets the size of each background image.	One or two values, each `auto` or a non-negative length or percentage; or `cover` or `contain`; comma-separated list	`auto`	CSS Backgrounds 4, `background-size`
`background-tbd`	Sets background layer properties without setting the background color.	One or more background layers	See individual properties	CSS Backgrounds 4, `background-tbd`
`isolation`	Controls whether an element forms an isolated group for blending.	`auto` or `isolate`	`auto`	Compositing and Blending 1, `isolation`
`mix-blend-mode`	Sets how an element blends with the content behind it.	`normal`, `darken`, `multiply`, `color-burn`, `lighten`, `screen`, `color-dodge`, `overlay`, `soft-light`, `hard-light`, `difference`, `exclusion`, hue, `saturation`, `color`, or `luminosity`	`normal`	Compositing and Blending 1, `mix-blend-mode`

- `<color>` means a CSS color, such as navy, `#336699`, or `rgb(51 102 153)`.
- `<image>` means none, an image such as `url("paper.png")`, or a gradient such as `linear-gradient(white, gray)`.

Display and positioning

The `display` property determines the kind of box an element generates. Positioning properties describe how a box is placed relative to its normal position or containing block.

Name	Description	Values	Default	Specification
`display`	Selects whether an element generates a box and describes its outer	none, `contents`, or combinations of outer types (`block`, `inline`, `run-in`), inner types		CSS Display 4 §2

Name	Description	Values	Default	Specification
	and inner display types.	(`flow`, `flow-root`, `table`, `flex`, `grid`, `ruby`), and `list-item`; also internal table and ruby types, legacy combinations such as `inline-block`, `inline-flex`, and `inline-grid`, plus `math`, `grid-lanes`, and `inline-grid-lanes`		
`visibility`	Controls whether a box is visible while preserving its place in layout.	`visible`, `hidden`, `force-hidden`, or `collapse`		CSS Display 4 §5
`order`	Changes the order of flex and grid items for layout and painting.	An integer		CSS Display 4 §3
`reading-flow`	Chooses the reading order of children in a block, flex, or grid container.	`normal`, `source-order`, `flex-visual`, `flex-flow`, `grid-rows`, `grid-columns`, or `grid-order`		CSS Display 4 §4
`reading-order`	Assigns an integer ordering value to a direct child of a reading-flow container.	An integer		CSS Display 4 §4.1
`position`	Chooses the positioning scheme for an element.	`static`, `relative`, `absolute`, `sticky`, `fixed`, or `running()`		CSS Position 3 §2; CSS GCPM 3 §1.2.1
`top`, `right`, `bottom`, `left`	Set an inset from the corresponding side of the containing block for a positioned element.	`auto`, a length or percentage, `anchor()`, or `anchor-size()`		CSS Position 3 §3.1
`inset`	Sets the four physical insets.	One to four values accepted by `top`, `right`, `bottom`, and `left`		CSS Position 3 §3.2

Name	Description	Values	Default	Specification
`inset-block`, `inset-inline`	Set the start and end insets along the block or inline axis.	One or two values accepted by the corresponding logical inset properties		CSS Position 3 §3.2
`inset-block-start`, `inset-block-end`, `inset-inline-start`, `inset-inline-end`	Set one inset on the block or inline axis.	`auto`, a length or percentage, `anchor()`, or `anchor-size()`		CSS Position 3 §3.1
`z-index`	Sets a positioned box's stack level.	`auto` or an integer		CSS 2 §9.9
`float`	Takes a box out of the normal flow and places it at the side, at the top or at the bottom of its container. The value `footnote` makes the element a footnote.	`none`, `left`, `right`, `top`, `bottom`, `block-start`, `block-end`, `inline-start`, `inline-end`, `snap-block`, `snap-inline` or `footnote`	`none`	CSS Page Floats 3 §3.2

- A length is a size with a unit, such as `12pt` or `1em`. A percentage is relative to the containing block, as defined for the relevant inset.
- `anchor()` uses an anchor's side or center as an inset coordinate. `anchor-size()` uses a selected anchor's size. See CSS Anchor Positioning 1 and the `anchor-size()` function.
- `running()` names an element to move into a page margin box. See CSS GCPM 3.
- An integer is a whole number without a fractional part. Negative values, zero, and positive values are allowed for `order` and `z-index`; CSS Display defines how ordering is applied.

Page reference

Name	Description	Values	Default	Specification
`page`	Assigns a named page type to a box.	`auto` or a custom identifier	`auto`	CSS Paged Media 3, page
`@page`	Defines page-box declarations and may contain page-margin rules.	Optional comma-separated page selectors		CSS Paged Media 3, @page

Name	Description	Values	Default	Specification
`size`	Sets page-box size and orientation. `auto` uses A4.	`auto`; one or two lengths; or A3, A4, A5, B4, B5, `JIS-B4`, `JIS-B5`, `ledger`, `legal`, or `letter`, optionally combined with `portrait` or `landscape`	A4	CSS Paged Media 3, `size`
`bleed`	Sets the bleed extension beyond the page box. `auto` uses `6pt` with crop marks and `0` otherwise.	`auto` or a length	`0` (`auto`)	CSS Paged Media 3, `bleed`
`marks`	Requests crop marks, registration marks, or both outside the page box.	none, `crop`, `cross`, or both `crop` and `cross`	none	CSS Paged Media 3, `marks`
`page-orientation`	Rotates the laid-out page in the output.	`upright`, `rotate-left`, or `rotate-right`	`upright`	CSS Paged Media 3, `page-orientation`
`@top-left-corner`	Defines the fixed-size box at the intersection of the top and left page margins.			CSS Paged Media 3, `@top-left-corner`
`@top-left`	Defines the box between the top-left corner and top-center boxes.			CSS Paged Media 3, `@top-left`
`@top-center`	Defines the centered box in the top page margin.			CSS Paged Media 3, `@top-center`
`@top-right`	Defines the box between the top-center and top-right corner boxes.			CSS Paged Media 3, `@top-right`

Name	Description	Values	Default	Specification
`@top-right-corner`	Defines the fixed-size box at the intersection of the top and right page margins.			CSS Paged Media 3, `@top-right-corner`
`@right-top`	Defines the box between the top-right corner and right-middle boxes.			CSS Paged Media 3, `@right-top`
`@right-middle`	Defines the centered box in the right page margin.			CSS Paged Media 3, `@right-middle`
`@right-bottom`	Defines the box between the right-middle and bottom-right corner boxes.			CSS Paged Media 3, `@right-bottom`
`@bottom-right-corner`	Defines the fixed-size box at the intersection of the bottom and right page margins.			CSS Paged Media 3, `@bottom-right-corner`
`@bottom-right`	Defines the box between the bottom-center and bottom-right corner boxes.			CSS Paged Media 3, `@bottom-right`
`@bottom-center`	Defines the centered box in the bottom page margin.			CSS Paged Media 3, `@bottom-center`
`@bottom-left`	Defines the box between the bottom-left corner and bottom-center boxes.			CSS Paged Media 3, `@bottom-left`
`@bottom-left-corner`	Defines the fixed-size box at the intersection of the bottom and left page margins.			CSS Paged Media 3, `@bottom-left-corner`

Name	Description	Values	Default	Specification
`@left-bottom`	Defines the box between the left-middle and bottom-left corner boxes.			CSS Paged Media 3, `@left-bottom`
`@left-middle`	Defines the centered box in the left page margin.			CSS Paged Media 3, `@left-middle`
`@left-top`	Defines the box between the top-left corner and left-middle boxes.			CSS Paged Media 3, `@left-top`
`@footnote`	Styles the footnote area at the bottom of the page. It goes inside an @page rule.	A block of declarations, such as borders, margins, padding and `max-height`.		CSS GCPM 3 §2.4

Declarations in page-margin rules

Each page-margin at-rule contains declarations written as `property: value;`. Use properties that apply to page-margin boxes, including `content`, font and text properties, color, backgrounds, borders, margins, padding, dimensions and `vertical-align`. See the CSS Paged Media page-margin property list.

For example, set a running title in the top-center box and a page number in the bottom-center box:

```
@page {
    @top-center {
        content: "Annual report";
        font-weight: bold;
    }

    @bottom-center {
        content: "Page " counter(page);
        text-align: center;
    }
}
```

Links and generated content reference

The values list the syntax defined by the cited specifications.

Name	Description	Values	Default	Specification
`content`	Replaces an element's contents or supplies generated content for pseudo-elements and page margin boxes.	`normal`, none, an image that replaces the contents, a list of generated items, or `element()`; a list can include alternative text after `/`		CSS Content 3 §1
`bookmark-label`	Sets the text for a generated bookmark.	A list of generated content items		CSS Content 3 §3.2
`bookmark-level`	Sets a bookmark's hierarchy level, or disables its creation.	none or an integer of 1 or greater		CSS Content 3 §3.1
`bookmark-state`	Sets whether a generated bookmark starts open or closed.	open or `closed`		CSS Content 3 §3.3
`quotes`	Selects the quotation marks used by generated `open-quote` and `close-quote` content.	`auto`, none, `match-parent`, or one or more pairs of quoted strings		CSS Content 3 §2.4.1
`string-set`	Assigns named values from an element for retrieval in page content.	none or comma-separated pairs of a name and one or more strings		CSS Content 3 §2.7.1
`string()`	Gives the value of a named string saved with `string-set`. Use it in the `content` property of a page-margin box.	A name, and an optional mode: `first`, `start`, `last` or `first-except`	`first`	CSS GCPM 3 §1.1.2
`content()`	Gives text of an element. Use it in the value of `string-set`.	`text`, `before`, `after` or `first-letter`	`text`	CSS GCPM 3 §1.1.1.1
`-cssmedia-link-counter()`	Gives the value of a counter at the destination of the link of the element, such as the number of a footnote that the link points to. CSSMedia extension.	The name of a counter, and an optional counter style.		CSSMedia extension

Name	Description	Values	Default	Specification
`-cssmedia-link`	Makes an XML element a link to another part of the document or to an external address. The values are joined in order to make the destination.	none, or one or more strings or `url()` values	none	CSSMedia extension
`link-parameters`	Supplies parameters for links to external resources.	none or a comma-separated list of `param()` values		CSS Link Parameters 1 §2.1

- A `<content-list>` is one or more generated content items, such as strings, quote keywords, counters, images, attribute values, or `content()` values. See CSS Content 3 and Named strings.
- A `<content-replacement>` is one image that replaces the element's normal contents. See CSS Content 3.
- The image form of `element()` uses an element ID to produce an image; the running-element form selects a named element for use in page margin boxes. See CSS Images 4 and CSS GCPM 3.
- A positive `<integer>` for `bookmark-level` is a whole number of at least 1. The top level is 1.
- A `<string>` is text between quotes. `url()` and `src()` provide URL values.
- A `<custom-ident>` is an author-chosen identifier, not a CSS-wide keyword or none.
- `param()` takes a dashed parameter name beginning with `--` and an optional `<declaration-value>`. If omitted, the parameter value is empty. A `<declaration-value>` is the sequence of CSS component values after a declaration's colon, excluding top-level semicolons and unmatched closing braces. See CSS Link Parameters 1 and CSS Syntax 3.

For examples of destinations, named strings, and generated content, see Links and Named strings.

Counters, list markers, and footnotes

Name	Description	Values	Default	Specification
`counter-reset`	Creates a new counter instance and initializes it.	none or one or more counter names, each optionally followed by an integer; the default value for each name is `0`		CSS Lists 3 §4.1

Name	Description	Values	Default	Specification
`counter-increment`	Increases each named counter, creating it if needed.	none or one or more counter names, each optionally followed by an integer; the default increment is `1`		CSS Lists 3 §4.2
`counter-set`	Sets the value of existing counters, creating a counter if it does not exist.	none or one or more counter names, each optionally followed by an integer; the default value is `0`		CSS Lists 3 §4.3
`list-style`	Sets the marker's position, image, and type.	Any order of `inside` or `outside`, an image or none, and a counter style, string, or none; any component can be omitted		CSS Lists 3 §3.6
`list-style-image`	Uses an image as the list marker.	An image or none		CSS Lists 3 §3.1
`list-style-position`	Places the marker inside the list item's contents or outside its principal box.	`inside` or `outside`		CSS Lists 3 §3.5
`list-style-type`	Chooses the marker's counter style or a literal marker string.	A counter-style name, a string, or none		CSS Lists 3 §3.4
`marker-side`	Chooses whether an outside marker's position follows the list item's directionality or its parent's.	`match-self` or `match-parent`		CSS Lists 3 §3.7
`@counter-style`	Defines a named counter style for markers and counter functions.	A name followed by counter-style descriptors		CSS Counter Styles 3 §3

Name	Description	Values	Default	Specification
`system` descriptor	Selects the algorithm that converts counter values to symbols.	`cyclic`, `numeric`, `alphabetic`, `symbolic`, `additive`, `fixed` with an optional starting integer, or `extends` with another counter-style name		CSS Counter Styles 3 §3.1
`symbols` descriptor	Lists the symbols used by the cyclic, fixed, symbolic, alphabetic, or numeric systems.	One or more symbols		CSS Counter Styles 3 §3.2
`additive-symbols` descriptor	Maps descending counter weights to symbols for the additive system.	Comma-separated pairs of a non-negative integer and a symbol, in descending weight order		CSS Counter Styles 3 §3.3
`negative` descriptor	Adds symbols before and optionally after a negative counter representation.	One or two symbols		CSS Counter Styles 3 §3.4
`prefix` descriptor	Adds a symbol before each marker.	One symbol		CSS Counter Styles 3 §3.7
`suffix` descriptor	Adds a symbol after each marker.	One symbol		CSS Counter Styles 3 §3.7
`range` descriptor	Limits the values that the counter style can represent before fallback is used.	`auto` or comma-separated pairs of integer bounds or `infinite`		CSS Counter Styles 3 §3.5
`pad` descriptor	Pads short representations to a minimum length with a chosen symbol.	A non-negative integer and a symbol		CSS Counter Styles 3 §3.6
`fallback` descriptor	Names the counter style used when this style cannot represent a value.	A counter-style name		CSS Counter Styles 3 §3.8

Name	Description	Values	Default	Specification
`footnote` counter	The counter that numbers the footnotes. Every footnote adds 1 to it. Change it with `counter-reset`, `counter-set` and `counter-increment`, in an element rule to number by section or in an `@page` rule to number by page.	Show it with `counter(footnote)`, with an optional counter style.	0 at the start of the document.	CSS GCPM 3 §2.5
`footnote-display`	Sets how a footnote body appears in the footnote area.	`block`, `inline`, or `compact`	`block`	CSS GCPM 3 §2.3
`footnote-policy`	Sets what happens when a footnote does not fit on the page of its call. `auto` continues it on the next pages, `line` moves the line with the call to the next page, and `block` moves the paragraph.	`auto`, `line`, or `block`	`auto`	CSS GCPM 3 §2.8

- A `<counter-name>` is an identifier naming a CSS counter. none and CSS-wide keywords are not counter names. See CSS Lists 3.
- An `<integer>` is a whole number without a fractional part. In the `range` descriptor, `infinite` means there is no bound in that direction. See CSS Values 4.
- An `<image>` is a URL, image function, or gradient that can provide image data. See CSS Images 3.
- A `<counter-style>` is a predefined counter style name, a name defined by `@counter-style`, or a `symbols()` function. See CSS Counter Styles 3.
- A `<counter-style-name>` is the identifier naming a predefined or author-defined counter style. See CSS Counter Styles 3.
- A `<symbol>` is a string, an identifier whose characters form the symbol, or an image. See CSS Counter Styles 3.

For example, this rule defines a two-symbol cycle. Replace `item` with an element name in your XML. The `display` declaration makes those elements list items, and their markers use the defined style.

```
@counter-style stars {
    system: cyclic;
    symbols: "★" "□";
    suffix: " ";
}

item {
    display: list-item;
    list-style-type: stars;
}
```

CSSMedia applies counter operations and prepares text marker content while building the box tree.

Footnotes

This reference lists the properties, the rule, the counter and the pseudo-elements that CSSMedia uses for footnotes.

Properties, rule and counter

Name	Description	Values	Default	Specification
`float`	Takes a box out of the normal flow and places it at the side, at the top or at the bottom of its container. The value `footnote` makes the element a footnote.	`none`, `left`, `right`, `top`, `bottom`, `block-start`, `block-end`, `inline-start`, `inline-end`, `snap-block`, `snap-inline` or `footnote`	`none`	CSS Page Floats 3 §3.2
`footnote-display`	Sets how a footnote body appears in the footnote area.	`block`, `inline`, or `compact`	`block`	CSS GCPM 3 §2.3
`footnote-policy`	Sets what happens when a footnote does not fit on the page of its call. `auto` continues it on the next pages, `line` moves the line with the call to the next page, and `block` moves the paragraph.	`auto`, `line`, or `block`	`auto`	CSS GCPM 3 §2.8

Name	Description	Values	Default	Specification
`footnote` counter	The counter that numbers the footnotes. Every footnote adds 1 to it. Change it with `counter-reset`, `counter-set` and `counter-increment`, in an element rule to number by section or in an `@page` rule to number by page.	Show it with `counter(footnote)`, with an optional counter style.	0 at the start of the document.	CSS GCPM 3 §2.5
`@footnote`	Styles the footnote area at the bottom of the page. It goes inside an `@page` rule.	A block of declarations, such as borders, margins, padding and `max-height`.		CSS GCPM 3 §2.4

Pseudo-elements

Name	Description	Values	In a PDF	Specification
`::footnote-call`	The reference that replaces a footnote in the text. By default it is the value of the `footnote` counter, as a superscript.		Matches	CSS GCPM 3 §2.6
`::footnote-marker`	The marker at the start of a footnote body. By default it is the value of the `footnote` counter, followed by a period and a space.		Matches	CSS GCPM 3 §2.7

Links

This reference lists the property, the function and the pseudo-classes that CSSMedia uses for links.

Property and function

Name	Description	Values	Default	Specification
`-cssmedia-link`	Makes an XML element a link to another part of the document or to an external address. The values are joined in order to make the destination.	none, or one or more strings or `url()` values	none	CSSMedia extension
`-cssmedia-link-counter()`	Gives the value of a counter at the destination of the link of the element, such as the number of a footnote that the link points to. CSSMedia extension.	The name of a counter, and an optional counter style.		CSSMedia extension

Pseudo-classes

Name	Description	Values	In a PDF	Specification
`:link`, `:any-link`	Elements that are hyperlinks, because `-cssmedia-link` gives them a destination.		Matches	Selectors 4 §8.1, §8.2
`:local-link`	Hyperlinks that point to the same document. The optional number limits how many path segments are compared.	An optional non-negative integer in parentheses.	Matches	Selectors 5 §2.1
`:visited`, `:target`	Links that were visited, and the element that a link points to. A PDF has no browsing history and no current target.		Never matches	Selectors 4 §8.2, §8.3

Named strings

This reference lists the property and the functions that CSSMedia uses for named strings.

Name	Description	Values	Default	Specification
`string-set`	Assigns named values from an element for retrieval in page content.	none or comma-separated pairs of a name and one or more strings		CSS Content 3 §2.7.1
`content()`	Gives text of an element. Use it in the value of `string-set`.	`text`, `before`, `after` or `first-letter`	`text`	CSS GCPM 3 §1.1.1.1
`string()`	Gives the value of a named string saved with `string-set`. Use it in the `content` property of a page-margin box.	A name, and an optional mode: `first`, `start`, `last` or `first-except`	`first`	CSS GCPM 3 §1.1.2

Cascade reference

The `Default` column lists values used by CSSMedia's current PDF-oriented rule evaluation. Media-query width, height and ratio values use the A4 default page size.

Property and at-rule reference

Name	Description	Values	Default	Specification
`all`	Resets all properties except `direction` and `unicode-bidi`.	`initial` | `inherit` | `unset` | `revert` | `revert-layer`		CSS Cascade 5 §3.2
`@charset`	Declares the stylesheet's fallback character encoding.	`@charset "<label>";`		CSS Syntax 3 §8.3
`@import`	Loads another stylesheet, optionally conditioned on a layer, support test or media query.	`@import <url-or-string> [layer | layer(<layer-name>)]? [supports(<condition>)]? <media-query-list>?;`		CSS Cascade 5 §2

Name	Description	Values	Default	Specification
`@namespace`	Declares a namespace prefix for selectors.	`@namespace <namespace-prefix>? <string-or-url>;`		CSS Namespaces 3 §3
`@layer`	Declares layer order or groups rules in a cascade layer.	`@layer <layer-name>? { <stylesheet> }` or `@layer <layer-name-list>;`		CSS Cascade 5 §6.4
`@property`	Registers a custom property with syntax, inheritance and optional initial value.	`@property <custom-property-name> { <declaration-list> }`		CSS Properties and Values API 1 §3
`syntax`	Defines the grammar of a registered custom property.	`<string>`		CSS Properties and Values API 1 §3.1
`inherits`	Sets whether a registered custom property inherits by default.	`true` | `false`		CSS Properties and Values API 1 §3.2
`initial-value`	Sets the registered custom property's initial value.	`<declaration-value>`	Guaranteed-invalid for `syntax: "*"`; otherwise required	CSS Properties and Values API 1 §3.3
`@media`	Includes rules when a media query matches the PDF environment.	`@media <media-query-list> { <rule-list> }`		CSS Conditional 3 §5

Name	Description	Values	Default	Specification
`@supports`	Includes rules when CSSMedia accepts a declaration, selector or supported feature test.	`@supports <supports-condition> { <rule-list> }`		CSS Conditional 3 §6
`@supports-condition`	Defines a named group of support tests.	`@supports-condition <supports-condition-name> { <block-contents> }`		CSS Conditional 5 §8
`@container`	Includes rules when a selected ancestor container passes the query.	`@container <container-condition># { <block-contents> }`		CSS Conditional 5 §5.4
`@scope`	Limits the selectors in a rule list to a scope root and optional scope limit.	`@scope [(<scope-start>)]? [to (<scope-end>)]? { <rule-list> }`		CSS Cascade 6 §2.5
`@custom-media`	Gives a name to a media query list or boolean value.	`@custom-media <extension-name> [<media-query-list> | true | false];`		Media Queries 5 §10
`@when`	Includes rules when a generalized boolean condition is true.	`@when <boolean-condition> { <rule-list> }`		CSS Conditional 5 §3
`@else`	Supplies the next branch of a conditional chain.	`@else <boolean-condition>? { <rule-list> }`		CSS Conditional 5 §4

Name	Description	Values	Default	Specification
`@function`	Defines a custom CSS function.	`@function <name>(<parameters>) [returns <type>]? { <declaration-rule-list> }`		CSS Functions and Mixins 1 §2.1
`result`	Supplies a custom function's return value.	`<declaration-value>?`		CSS Functions and Mixins 1 §2.2

- ? means optional, | means choose one, # means one or more comma-separated items, || means one or more options in any order, and brackets group parts of the syntax.
- `<label>` is an encoding name.
- `<url-or-string>` and `<string-or-url>` each mean a URL or quoted string.
- `<namespace-prefix>` is the prefix bound to a namespace.
- `<layer-name>` names a cascade layer. `<layer-name-list>` is a comma-separated list of layer names.
- `<stylesheet>` is a CSS stylesheet.
- `<custom-property-name>` is a custom property name beginning with `--`.
- `<declaration-list>` is a list of declarations.
- `<block-contents>` is content inside a rule's braces, including declarations and nested rules.
- `<declaration-value>` is the value part of a declaration: one or more CSS tokens, such as `navy`, `calc(1em + 2px)` or `var(--size)`. It cannot contain a top-level semicolon or `!`, since those delimit declarations and `!important`. Semicolons and exclamation marks inside functions are not top-level. See CSS Syntax 3.
- `<string>` is a quoted CSS string.
- `<media-query-list>` is one or more comma-separated media queries.
- `<rule-list>` is a sequence of rules.
- `<condition>` and `<supports-condition>` are support conditions made from feature tests and boolean operators.
- `<supports-condition-name>` names a reusable support condition.
- `<container-condition>` optionally names a query container and gives its query.
- `<boolean-condition>` combines `media()` and `supports()` tests.
- `<scope-start>` and `<scope-end>` are selector lists that set the scope roots and limits of `@scope`.
- `<extension-name>` names a custom media query.
- `<name>` in the function syntax is a function name beginning with `--`.
- `<parameters>` is a comma-separated list of function parameters.
- `<type>` is the optional syntax expected for the function result.

- `<declaration-rule-list>` contains declarations and nested rules.
- `<number>` in the function example is a CSS number, which can be an integer or include a fractional part.

The `@charset` rule must be the first rule in a stylesheet to affect byte decoding. Use `@namespace` before ordinary rules, and place `@import` rules before namespace declarations and ordinary rules.

Function names start with `--`. Parameters can have a syntax type and a default value. Use the `result` descriptor for the returned value:

```
@function --double(--value <number>) returns <number> {
    result: calc(var(--value) * 2);
}
```

`@supports` declaration tests use parentheses. `selector()` tests one complex selector. `font-tech()`, `font-format()` and `at-rule()` test supported features. A named condition is tested by placing its name inside parentheses:

```
@supports (color: navy) and selector(chapter > title) {
    chapter > title { color: navy; }
}

@supports-condition --has-navy {
    color: navy;
}

@supports (--has-navy) {
    chapter { color: navy; }
}
```

`@when` conditions combine `media()` and `supports()` tests with `and`, `or` and `not`. Parentheses group mixed operators:

```
@when media(width >= 6in) and supports(color: navy) {
    chapter { color: navy; }
}
```

Use `@else` immediately after an `@when`, `@media`, `@supports` or `@container` branch. An unrelated rule ends the chain.

Media features

Media feature values describe the page or PDF environment used for conditional rule evaluation. They do not describe a screen.

Name	Description	Values	Default	Specification
`any-hover`	Whether any pointing device can hover.	`none` | `hover`	`none`	Media Queries 5
`any-pointer`	The accuracy of any available pointing device.	`none` | `coarse` | `fine`	`none`	Media Queries 5

Name	Description	Values	Default	Specification
`aspect-ratio`	The ratio of page width to page height.	`<ratio>`	A4 page ratio	Media Queries 5
`color`	Bits per color component in the output.	`<integer>`	`8`	Media Queries 5
`color-gamut`	The approximate output color gamut.	`srgb` | `p3` | `rec2020`	`srgb`	Media Queries 5
`color-index`	Number of entries in the output color lookup table.	`<integer>`	`0`	Media Queries 5
`device-aspect-ratio`	The ratio of output width to output height.	`<ratio>`	A4 page ratio	Media Queries 5
`device-height`	Output height.	`<length>`	A4 page height	Media Queries 5
`device-width`	Output width.	`<length>`	A4 page width	Media Queries 5
`display-mode`	The presentation mode.	`fullscreen` | `standalone` | `minimal-ui` | `browser` | `picture-in-picture`	Not set	Media Queries 5
`dynamic-range`	The output's dynamic range.	`standard` | `high`	`standard`	Media Queries 5
`environment-blending`	Whether output blends with the surrounding environment.	`opaque` | `additive` | `subtractive`	`opaque`	Media Queries 5
`forced-colors`	Whether the output forces a limited color palette.	`none` | `active`	`none`	Media Queries 5
`grid`	Whether the output is grid-based.	`<mq-boolean>` (`0` or `1`)	`0`	Media Queries 5

Name	Description	Values	Default	Specification
`height`	Page height.	`<length>`	A4 page height	Media Queries 5
`horizontal-viewport-segments`	Number of horizontal viewport segments.	`<integer>`	`1`	Media Queries 5
`hover`	Whether the primary pointing device can hover.	`none` | `hover`	`none`	Media Queries 5
`inverted-colors`	Whether the output inverts colors.	`none` | `inverted`	`none`	Media Queries 5
`monochrome`	Number of bits per pixel in a monochrome output.	`<integer>`	`0`	Media Queries 5
`nav-controls`	Availability of navigation controls.	`none` | `back`	`none`	Media Queries 5
`orientation`	Whether the page is portrait or landscape.	`portrait` | `landscape`	`portrait`	Media Queries 5
`overflow-block`	How the output handles block-axis overflow.	`none` | `scroll` | `paged`	`paged`	Media Queries 5
`overflow-inline`	How the output handles inline-axis overflow.	`none` | `scroll`	`none`	Media Queries 5
`pointer`	The accuracy of the primary pointing device.	`none` | `coarse` | `fine`	`none`	Media Queries 5
`prefers-color-scheme`	The preferred color scheme.	`light` | `dark`	`light`	Media Queries 5
`prefers-contrast`	The preferred contrast level.	`no-preference` | `less` | `more` | `custom`	`no-preference`	Media Queries 5
`prefers-reduced-data`	Whether the user prefers reduced data use.	`no-preference` | `reduce`	`no-preference`	Media Queries 5

Name	Description	Values	Default	Specification
`prefers-reduced-motion`	Whether the user prefers reduced motion.	`no-preference` | `reduce`	`no-preference`	Media Queries 5
`prefers-reduced-transparency`	Whether the user prefers reduced transparency.	`no-preference` | `reduce`	`no-preference`	Media Queries 5
`resolution`	Output pixel density.	`<resolution>` | `infinite`	`infinite`	Media Queries 5
`scan`	The output's scanning process.	`interlace` | `progressive`	Not set	Media Queries 5
`scripting`	Whether scripting is available.	`none` | `initial-only` | `enabled`	`none`	Media Queries 5
`update`	How frequently the output can update.	`none` | `slow` | `fast`	`none`	Media Queries 5
`vertical-viewport-segments`	Number of vertical viewport segments.	`<integer>`	`1`	Media Queries 5
`video-color-gamut`	The color gamut of the output video.	`srgb` | `p3` | `rec2020`	`srgb`	Media Queries 5
`video-dynamic-range`	The dynamic range of the output video.	`standard` | `high`	`standard`	Media Queries 5
`width`	Page width.	`<length>`	A4 page width	Media Queries 5

- `<ratio>` is a ratio of two numbers.
- `<integer>` is a whole number.
- `<length>` is a CSS length, such as `12pt` or `2em`.
- `<mq-boolean>` is an integer with the value `0` or `1`.
- `<resolution>` is a pixel density, such as `300dpi`.

Container style queries

A container style query uses `style()` to test a computed style or custom property on an ancestor. See Conditional rules.

Glossary

CSS

Cascading Style Sheets (CSS) is the language, defined by the W3C, that describes how elements of a document are presented. A stylesheet is a list of rules. Each rule has a selector that chooses elements and declarations that set their properties. CSSMedia reads a CSS file to decide how to format an XML document in a PDF.

DTD

A Document Type Definition (DTD) is a grammar, defined by the XML specification, that declares the elements and attributes an XML document may use. A document points to its DTD in its document type declaration. When the DTD is not at the location the document names, an OASIS Catalog tells CSSMedia where to find it.

MathML

Mathematical Markup Language (MathML) is a W3C XML language for writing mathematical notation. A MathML formula describes the structure of an expression, such as fractions, roots and subscripts. CSSMedia draws MathML found in a document as an image.

Namespace

An XML namespace is a URI that identifies the vocabulary an element or attribute belongs to. An XML document declares it with an `xmlns` attribute, so elements with the same name in different vocabularies do not collide.

OASIS Catalog

An OASIS Catalog is an XML file, defined by the OASIS XML Catalogs standard, that maps the identifiers used in documents to the locations where the files are stored. A catalog lets a parser find a DTD, an XML Schema or a RelaxNG schema without downloading it. Pass a catalog to CSSMedia with the `-g` option.

PDF

Portable Document Format (PDF) is the file format, standardized as ISO 32000, for documents that look the same on any device. A PDF holds pages with text, fonts, images and vector graphics at fixed positions, and may also hold links and bookmarks. CSSMedia writes a PDF file from an XML file and a CSS file.

RelaxNG

RelaxNG is a schema language for XML, standardized as ISO/IEC 19757-2. A RelaxNG schema describes the structure of a document in an XML syntax or in a compact non-XML syntax. It is an alternative to a DTD and to an XML Schema, and it can be located with an OASIS Catalog.

SVG

Scalable Vector Graphics (SVG) is a W3C format for two-dimensional vector graphics, written as XML. An SVG drawing describes shapes, paths, text and images that can be scaled without losing quality. CSSMedia draws SVG found in a document as an image.

XML

Extensible Markup Language (XML) is a W3C standard for writing structured documents. An XML document is a tree of elements, with attributes and text, whose names are chosen by the author or by a vocabulary such as DITA or DocBook. CSSMedia reads an XML file as the content of the PDF.

XML Schema

XML Schema is a W3C language for describing the structure of XML documents and the data types of their content. A schema is itself an XML file, and it is more expressive than a DTD. Like a DTD, a schema can be located with an OASIS Catalog.